

EE 2101: Quiz 2

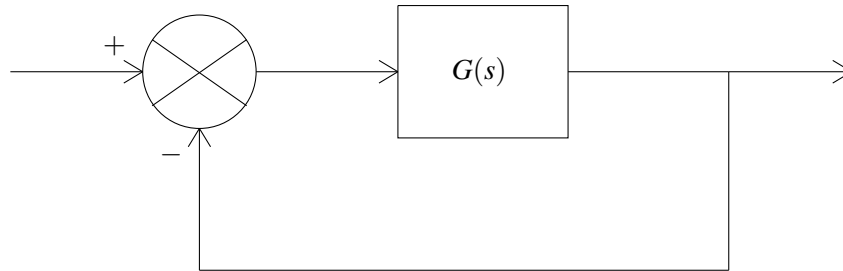


Figure 1: Control System

1. Consider the control system shown in Figure 1 with

[Marks: 2]

$$G(s) = \frac{100}{s(s+1)(s+5)^2}$$

Let $u(t)$ indicate the unit-step signal. Find the steady-state error corresponding to the following inputs:

- (a) $10u(t)$ (b) $10tu(t)$

2. Consider the control system shown in Figure 1 with

[Marks: 3]

$$G(s) = \frac{K}{s(s+7)^2}$$

The unit step response of this system has 25% overshoot. Find all closed-loop poles of the system.

3. Consider the control system shown in Figure 1 with

[Marks: 6]

$$G(s) = \frac{K(s+6)}{(s^2+10s+41)(s-2)(s+4)}$$

- (a) Plot the locus of closed-loop poles of the system as K is varied from 0 to ∞ . Show the asymptotes and the angles of arrival/departure.
- (b) Find the break-off point.
- (c) The output of this system corresponding to input signal $\sin(w_1 t)$ is unbounded. Find w_1 .

4. Consider an uncompensated system shown in Figure 1, that is operating at 30% overshoot and has

$$G(s) = \frac{K}{(s^2 + 6s + 13)(s + 15)}$$

The compensated system is required to have the following specifications in the unit step response:

- (a) 30% overshoot and (b) half the settling time as that of the uncompensated system

Design a cascade compensator with a zero at $s = -7$ to achieve these specifications. **[Marks: 6]**

5. Consider the control system shown in Figure 1. Derive the formula for the intersection of asymptotes in the root locus of this system. Note that just writing the formula will get zero marks. **[Marks: 3]**

All the Best!